

TAKE-HOME QUIZ 4

MATH 32A

Note: Due March 15 Saturday at 11:59pm

There are three questions in total.

1. Let $x = s + t$ and $y = s - t$. Show that for any differentiable function $f(x, y)$,

$$\left(\frac{\partial f}{\partial x}\right)^2 - \left(\frac{\partial f}{\partial y}\right)^2 = \frac{\partial f}{\partial s} \frac{\partial f}{\partial t}.$$

2. Find the local maximum and minimum values and saddle points of $f(x, y) = x^2 + xy + y^2 + y$.

3. A rectangular box without a lid is to be made from 12 m^2 of cardboard. Find the maximum volume of such a box.